Lesson 13: Some Important Continuous RVs

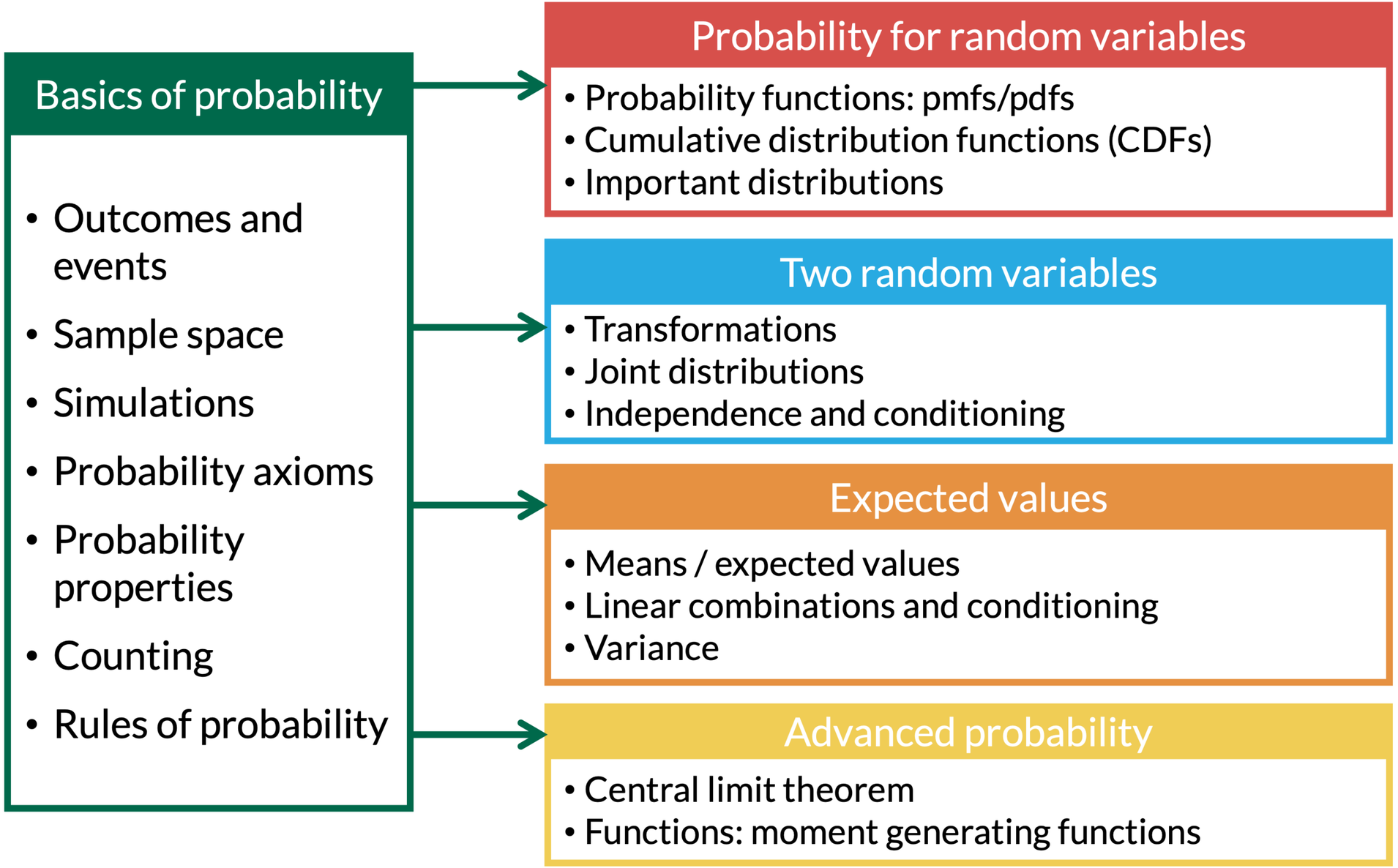
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Learning Objectives

1. Distinguish between Uniform, Exponential, Gamma, and Normal distributions when reading a word problem.
2. Identify the variable and the parameters in a word problem, and state what the variable and parameters mean.
3. Use the formulas for the pdf/CDF, expected value, and variance to answer questions and find probabilities.

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

Continuous uniform RVs

Properties of continuous uniform RVs

- **Scenario:** Events are **equally likely** to happen anywhere or anytime in an interval of values
- Shorthand: $X \sim \text{U}[a, b]$

X

X = a value in the interval $[a, b]$, where every value is equally likely

CDF

$$F_X(x) = \int_a^x f_X(s)\,ds = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \le x \le b \\ 1 & x > b \end{cases}$$

$\text{E}(X)$

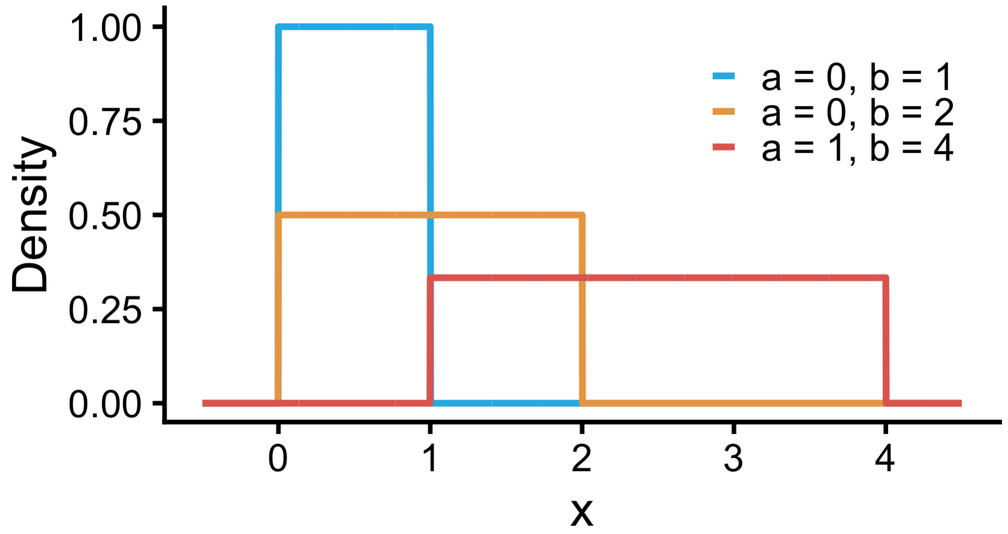
$$\text{E}(X) = \int_a^b x f_X(x)\,dx = \frac{a+b}{2}$$

$\text{Var}(X)$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

pdf

$$f_X(x) = \frac{1}{b-a}, \text{ for } a \le x \le b$$



Identifying continuous uniform RV from word problems

- Look for some indication that all events are **equally likely**
  - Could also say “uniformly distributed”
- Look for an interval
  - **Time example:** Customer in your store will approach the cash register in next 30 minutes. Approaching the register throughout the 30 minutes is equally likely.
  - **Length example:** You have a 12 inch string that you need to cut. You are equally likely to cut anywhere on the string.
- Different than the **discrete** uniform
  - Discrete usually includes a *countable* number of events that are equally likely
  - Continuous is not countable
    - Exact time and length can be measured with infinite decimal places

Uniform distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`runif()`	returns sample of random variables with specified uniform distribution	Simulate a Uniform RV
`dunif()`	returns density $f_X(x)$ at a certain value	$f_X(x)$: the height of the pdf, **not** a probability
`punif()`	returns cumulative probability of getting a certain value or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qunif()`	returns value corresponding to desired quantile	Inverse CDF: we give it a probability p and it returns the value t where $P(X \leq t) = p$

- Parameters: `min` $= a$ and `max` $= b$

Arrival times

Example 1

People arrive at equally likely times between 10 am and 11 am. Let X = minutes after 10 am so $X \sim \text{U}[0, 60]$

a. What is the probability that someone arrives at 10:30 am or earlier?

b. Find the time t where the probability of arriving at t or earlier is 0.35.

c. What is the probability that someone arrives between 10:14 and 10:16 am?

a. $P(X \leq 30)$

```
punif(q = 30, min = 0, max = 60)
```

```
[1] 0.5
```

b. Find t with $P(X \leq t) = 0.35$

```
qunif(p = 0.35, min = 0, max = 60)
```

```
[1] 21
```

c. $P(14 \leq X \leq 16)$

```
punif(q = 16, min = 0, max = 60) -
  punif(q = 14, min = 0, max = 60)
```

```
[1] 0.03333333
```

Arrival times: simulate it

Example 1

Simulate 10,000 arrival times and answer parts a, b, and c.

```
1 set.seed(1109)                                                    ①
2 reps <- 10000                                                     ②
3
4 arrival_sims <- tibble(sim = 1:reps) |>                           ③
5   mutate(
6     X = runif(n(), min = 0, max = 60)                             ④
7   )
```

```
1 arrival_sims |>                                                   ①
2   summarise(
3     prob_a = mean(X <= 30),                                       ②
4
5     t_b    = quantile(X, 0.35),                                   ③
6
7     prob_c = mean(X >= 14 & X <= 16)                              ④
8
9   )
```

```
# A tibble: 1 × 3
  prob_a   t_b prob_c
   <dbl> <dbl>  <dbl>
1  0.497  21.5 0.0319
```

Exponential RVs

Properties of exponential RVs

- **Scenario:** Modeling the time until the next (first) event
- Continuous analog to the discrete geometric distribution!
- Shorthand: $X \sim \text{Exp}(\lambda)$
  - λ = the **rate**: average number of events (successes) per unit of time

X

X = time until the next (first) event

CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

$\text{E}(X)$

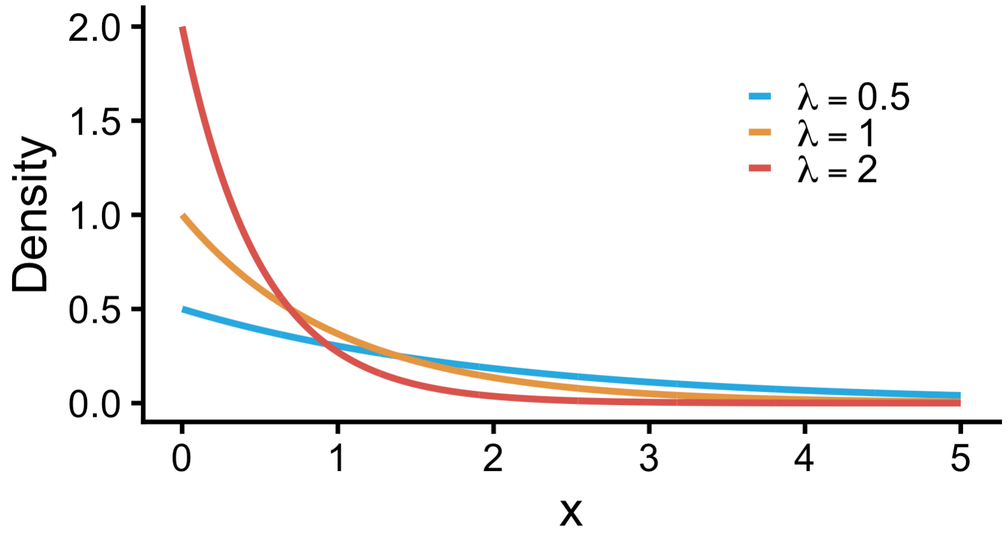
$$\text{E}(X) = \frac{1}{\lambda}$$

$\text{Var}(X)$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

pdf

$$f_X(x) = \lambda e^{-\lambda x} \text{ for } x > 0, \lambda > 0$$



Memoryless Property

If $b > 0$,

$$P(X > a + b | X > a) = P(X > b)$$

- **This can be interpreted as:**
  - If you have waited a seconds (or any other measure of time) without a success
  - Then the probability that you have to wait b more seconds is the same as the probability of waiting b seconds initially.

Identifying exponential RV from word problems

- Look for time between events/successes
- Look for a rate of the events over time period
- How does it differ from the geometric distribution?
  - Geometric is *number of trials* until first success
  - Exponential is *time* until first success
- Relation to discrete distributions
  - **Relationship to Geometric:** Exponential is the continuous analog to the Geometric distribution
  - **Relationship to Poisson:** When the time between arrivals is exponential, the number of arrivals in a fixed time interval is Poisson with the mean λ

Minimum of independent exponential RVs

Theorem 1

If $X_1, X_2, \dots, X_n$ are independent with $X_i \sim \text{Exp}(\lambda_i)$, then the **first** of the arrival times,

$$M = \min(X_1, X_2, \dots, X_n) \sim \text{Exp}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$$

- The rates add: waiting for the first of several independent events is like waiting for one event with the combined rate
- For example, if bus A comes every 10 minutes on average ($\lambda_1 = 1/10$) and bus B every 15 minutes ($\lambda_2 = 1/15$), the wait for whichever bus comes first is $\text{Exp}(1/10 + 1/15) = \text{Exp}(1/6)$: 6 minutes on average
- We can show this with the CDF method (Lesson 14)

Exponential distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rexp()`	returns sample of random variables with specified exponential distribution	Simulate a Exponential RV
`dexp()`	returns density $f_X(x)$ at a certain value	$f_X(x)$: the height of the pdf, **not** a probability
`pexp()`	returns cumulative probability of getting a certain value or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qexp()`	returns value corresponding to desired quantile	Inverse CDF: we give it a probability p and it returns the value t where $P(X \leq t) = p$

- Parameters: `rate` $= \lambda$

Waiting for the bus

Example 2

We are at the bus stop, measuring the time until our bus arrives. The bus comes every 10 minutes on average, so $X \sim \text{Exp}(\lambda = 1/10)$, with X in minutes.

a. What is the probability that the bus arrives in the next 5 minutes?

b. Find the time t where the probability of the bus arriving at t or earlier is 0.35.

c. What is the probability that the bus arrives between 3 and 5 minutes?

a. $P(X \leq 5)$

```
pexp(q = 5, rate = 1/10)
```

```
[1] 0.3934693
```

b. Find t with $P(X \leq t) = 0.35$

```
qexp(p = 0.35, rate = 1/10)
```

```
[1] 4.307829
```

c. $P(3 \leq X \leq 5)$

```
pexp(q = 5, rate = 1/10) -
  pexp(q = 3, rate = 1/10)
```

```
[1] 0.1342876
```

Waiting for the bus: simulate it

Example 2

Simulate 10,000 waits for the bus and answer parts a, b, and c.

```
1 bus_sims <- tibble(sim = 1:reps) |>                    ①
2   mutate(
3     X = rexp(n(), rate = 1/10)                         ②
4   )
```

```
1 bus_sims |>                                            ①
2   summarise(
3     prob_a = mean(X <= 5),                             ②
4
5     t_b    = quantile(X, 0.35),                        ③
6
7     prob_c = mean(X >= 3 & X <= 5)                     ④
8
9   )
```

```
# A tibble: 1 × 3
  prob_a   t_b prob_c
   <dbl> <dbl>  <dbl>
1  0.391  4.32  0.134
```

Memoryless property: check with a simulation

- We've already waited $a = 3$ minutes. Is the chance of waiting $b = 5$ **more** minutes the same as $P(X > 5)$?

```
1 bus_sims |>                                                    ①
2   summarise(
3     prob_over_5   = mean(X > 5),                               ②
4   )
```

```
# A tibble: 1 × 1
  prob_over_5
        <dbl>
1       0.609
```

Exact: both are $e^{-5/10} \approx 0.6065$

- At home: try other values of a, like `mean(X > 15) / mean(X > 10)`

Gamma RVs

Properties of gamma RVs

- **Scenario:** Modeling the time until the rth event.
- Shorthand: $X \sim \text{Gamma}(r, \lambda)$ or $X \sim \text{Gamma}(\alpha, \beta)$
  - Common to see $\alpha = r$ and $\beta = 1/\lambda$

X

X = time until the rth event

CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} \sum_{j=0}^{r-1} \frac{(\lambda x)^j}{j!} & x \geq 0 \end{cases}$$

$\text{E}(X)$

$$\text{E}(X) = \frac{r}{\lambda}$$

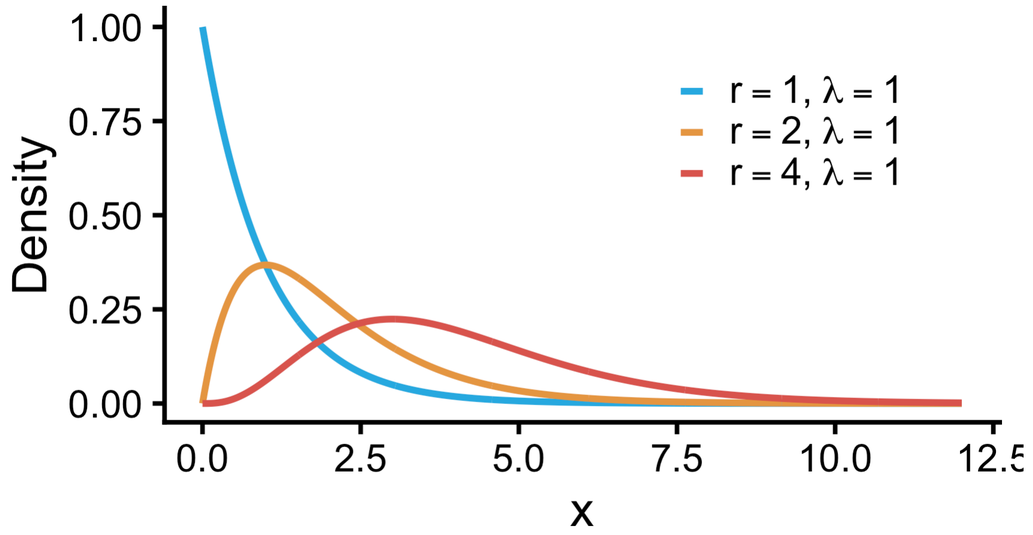
$\text{Var}(X)$

$$\text{Var}(X) = \frac{r}{\lambda^2}$$

pdf

$$f_X(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x} \text{ for } x > 0, \lambda > 0$$

where $\Gamma(r) = (r-1)!$



Identifying gamma RV from word problems

- Continuous analog to the Negative Binomial distribution
- Same as the sum of r independent waiting times: $X = \sum_{i=1}^{r} Y_i$, where $Y_i \sim \text{Exp}(\lambda)$
- Gamma distribution with $r = 1$ is same as exponential (both **continuous**)
  - Just like Negative Binomial with $r = 1$ is same as the geometric distribution (both **discrete**)
- Similar to exponential
  - Look for time between or until events/successes
    - BUT now we are measuring time until more than 1 success
  - Look for a rate of the events over time period

Gamma distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rgamma()`	returns sample of random variables with specified gamma distribution	Simulate a Gamma RV
`dgamma()`	returns density $f_X(x)$ at a certain value	$f_X(x)$: the height of the pdf, **not** a probability
`pgamma()`	returns cumulative probability of getting a certain value or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qgamma()`	returns value corresponding to desired quantile	Inverse CDF: we give it a probability p and it returns the value t where $P(X \leq t) = p$

- Parameters: `shape` $= r$ and `rate` $= \lambda$ (or `scale` $= \beta = 1/\lambda$)
- Another helpful site with R code: https://rpubs.com/mpfoley73/459051

Waiting for 4 buses

Example 3

We are at the bus stop, measuring the time until 4 buses arrive. The bus comes every 10 minutes on average, so $X \sim \text{Gamma}(r = 4, \lambda = 1/10)$, with X in minutes.

a. What is the probability that the 4 buses arrive in the next 50 minutes?

b. Find the time t where the probability of the 4 buses arriving at t or earlier is 0.35.

c. What is the probability that the 4 buses arrive between 30 and 50 minutes?

a. $P(X \leq 50)$

```
1 pgamma(q = 50, shape = 4, rate = 1/10)
```

```
[1] 0.7349741
```

b. Find t with $P(X \leq t) = 0.35$

```
1 qgamma(p = 0.35, shape = 4, rate = 1/10)
```

```
[1] 29.87645
```

c. $P(30 \leq X \leq 50)$

```
1 pgamma(q = 50, shape = 4, rate = 1/10) -
2   pgamma(q = 30, shape = 4, rate = 1/10)
```

```
[1] 0.382206
```

Waiting for 4 buses: simulate it

Example 3

Simulate 10,000 waits for 4 buses and answer parts a, b, and c.

```
1 n_buses <- 4                                                    ①
2
3 gamma_sims <- tibble(sim = 1:reps) |>                           ②
4   mutate(
5     X = rgamma(n(), shape = n_buses, rate=                      ③
6   )
```

```
1 gamma_sims |>                                                   ①
2   summarise(
3     prob_a = mean(X <= 50),                                     ②
4
5     t_b    = quantile(X, 0.35),                                 ③
6
7     prob_c = mean(X >= 30 & X <= 50)                            ④
8
9   )
```

```
# A tibble: 1 × 3
  prob_a   t_b prob_c
   <dbl> <dbl>  <dbl>
1  0.739  29.8  0.384
```

Remarks

- The parameter r in a Gamma(r,λ) distribution does NOT need to be a positive integer
  - r is usually a positive integer
- When r is a positive integer, the distribution is sometimes called an Erlang(r,λ) distribution
  - Erlang is a gamma with a whole-number r

- When r is any positive real number, we have a general gamma distribution that is usually instead parameterized by $\alpha > 0$ and $\beta > 0$, where:
  - $\alpha = \text{shape parameter}$: same as r, the total number of events we must witness
    - In Example 3: 4 buses to wait for (`shape = 4`)
  - $\beta = \text{scale parameter}$: same as $1/\lambda$, one over the rate parameter
    - In Example 3: 1 bus per 10 minutes, so `rate = 1/10` or `scale = 10`
    - In R: `rate = 1/scale`

Gamma as a sum of exponentials: simulate

1. Set up

- Is the gamma really a sum of exponentials? Wait for 4 buses, one after another, with each wait $Y_i \sim \text{Exp}(1/10)$. Is the total wait like the $X \sim \text{Gamma}(4, 1/10)$ from Example 3?

2. Simulate

```
1 waits <- expand_grid(sim = 1:reps, bus = 1:n_buses) |>
2   mutate(wait = rexp(n(), rate = 1/10))
3
4 sum_sims <- waits |>
5   group_by(sim) |>
6   summarise(Y = sum(wait))
```

①

②

```
1 head(sum_sims)
```

①

```
# A tibble: 6 × 2
    sim     Y
  <int> <dbl>
1     1  31.7
2     2  21.2
3     3  45.2
4     4  26.3
5     5  21.5
6     6  52.9
```

Gamma as a sum of exponentials: summarize

3. Summarize

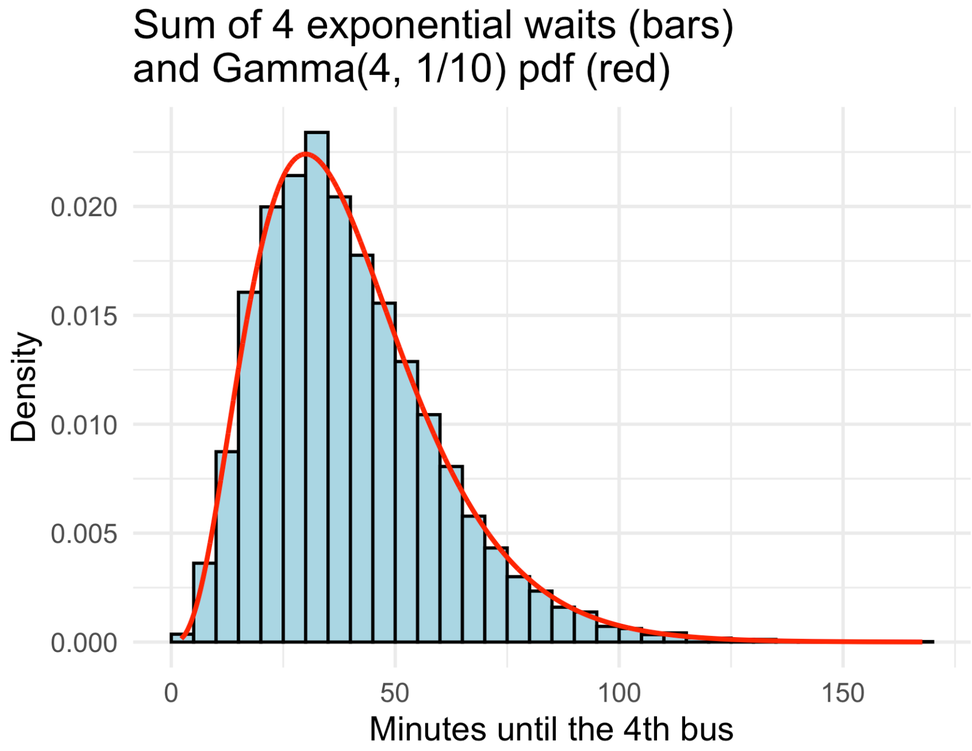
```
1  sum_sims |>                                  ①
2    summarise(
3      average = mean(Y),                       ②
4      std_dev = sd(Y),
5      prob_a  = mean(Y <= 50)                  ③
6    )
```

```
# A tibble: 1 × 3
  average std_dev prob_a
    <dbl>   <dbl>  <dbl>
1    39.9    19.9  0.737
```

4. Sensitivity analysis

- At home: change to `n_buses <- 1`. Does Y look like $\text{Exp}(1/10)$, with average 10?

▶ Plot code



Normal RVs

Properties of Normal RVs

- No scenario description here because the Normal distribution is so universal
  - Central Limit Theorem (Lesson 17) makes it applicable to many types of events
- Shorthand: $X \sim \text{Normal}(\mu, \sigma^2)$
  - The second parameter is the **variance** σ^2, but R's `*norm()` functions use `sd` $= \sigma$

E(X)

$$\text{E}(X) = \mu$$

Var(X)

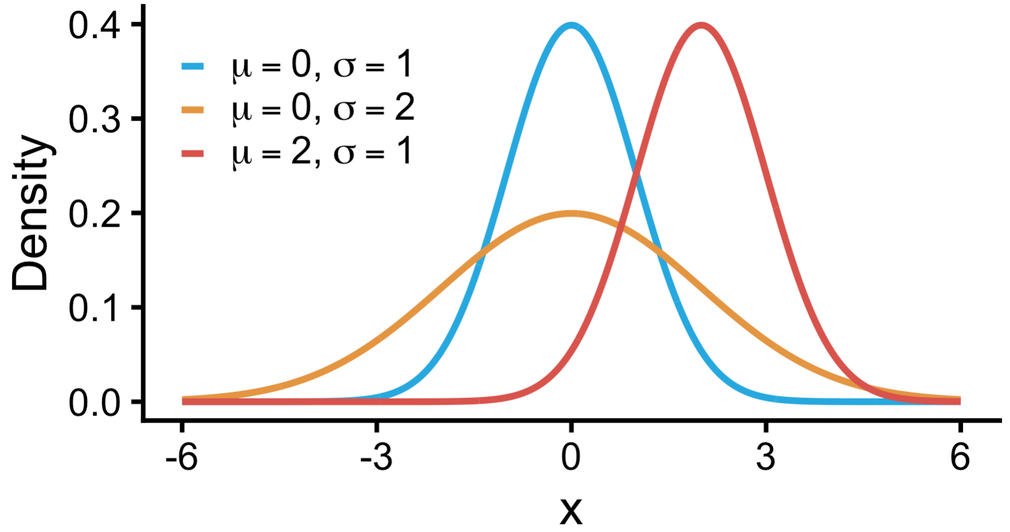
$$\text{Var}(X) = \sigma^2 \qquad \text{SD}(X) = \sigma$$

- No closed-form CDF: we use R (or a table) to find probabilities

pdf

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$$

for $-\infty < x < \infty$



Normal distribution: R commands

R commands with their input and output:

R code	What does it return?	What do we use it for?
`rnorm()`	returns sample of random variables with specified normal distribution	Simulate a Normal RV
`dnorm()`	returns density $f_X(x)$ at a certain value	$f_X(x)$: the height of the pdf, **not** a probability
`pnorm()`	returns cumulative probability of getting a certain value or less	$P(X \leq x)$ (or $P(X > x)$ if we add `lower.tail = FALSE`)
`qnorm()`	returns value corresponding to desired quantile	Inverse CDF: we give it a probability p and it returns the value t where $P(X \leq t) = p$

- Parameters: `mean` $= \mu$ and `sd` $= \sigma$ (the standard deviation, **not** the variance σ^2)

High temperature

Example 4

We are measuring the high temperature today. The average high temperature on this day across many, many years is 50 degrees with a standard deviation of 4 degrees, so $X \sim \text{Normal}(50, 4^2)$.

a. What is the probability that the high temperature is below 45 degrees?

b. Find the temperature t where the probability that the temperature is at t or lower is 0.35.

c. What is the probability that the temperature is between 45 and 50 degrees?

a. $P(X < 45)$

```
1 pnorm(q = 45, mean = 50, sd = 4)
```

```
[1] 0.1056498
```

b. Find t with $P(X \leq t) = 0.35$

```
1 qnorm(p = 0.35, mean = 50, sd = 4)
```

```
[1] 48.45872
```

c. $P(45 \leq X \leq 50)$

```
1 pnorm(q = 50, mean = 50, sd = 4) -
2   pnorm(q = 45, mean = 50, sd = 4)
```

```
[1] 0.3943502
```

High temperature: simulate it

Example 4

Simulate 10,000 high temperatures and answer parts a, b, and c.

```
1  temp_sims <- tibble(sim = 1:reps) |>                    ①
2    mutate(
3      X = rnorm(n(), mean = 50, sd = 4)                   ②
4    )
```

```
1  temp_sims |>                                            ①
2    summarise(
3      prob_a = mean(X < 45),                              ②
4
5      t_b    = quantile(X, 0.35),                         ③
6
7      prob_c = mean(X >= 45 & X <= 50)                    ④
8
9    )
```

```
# A tibble: 1 × 3
  prob_a   t_b prob_c
   <dbl> <dbl>  <dbl>
1  0.104  48.5  0.389
```

Standard Normal Distribution

$$Z \sim \text{Normal}(\mu = 0, \sigma^2 = 1)$$

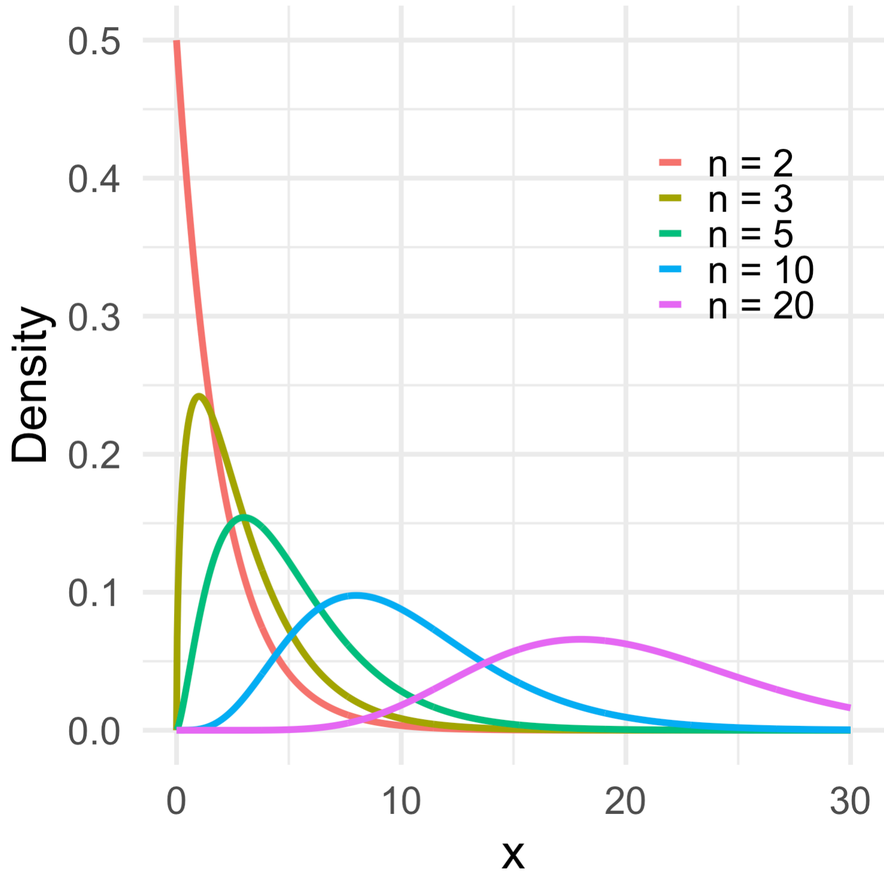
- Used to be more helpful when computing was not as advanced
  - Use tables of the standard normal
  - You can convert any normal distribution to a standard normal through transformation
- For $X \sim \text{Normal}(\mu_X, \sigma_X^2)$: $Z = \dfrac{X - \mu_X}{\sigma_X}$
  - Comes from $X = \sigma_X Z + \mu_X$
  - Since σ_X and μ_X are constants, then $E(X) = \sigma_X E(Z) + \mu_X = \mu_X$ and $SD(X) = \sigma_X$

Chi-squared distribution

- If $Z \sim \text{Normal}(0, 1)$, then $X = Z^2$ has a Chi-squared distribution with 1 degree of freedom
- If $Z_1, Z_2, \ldots, Z_n$ are independent standard normal RVs, then

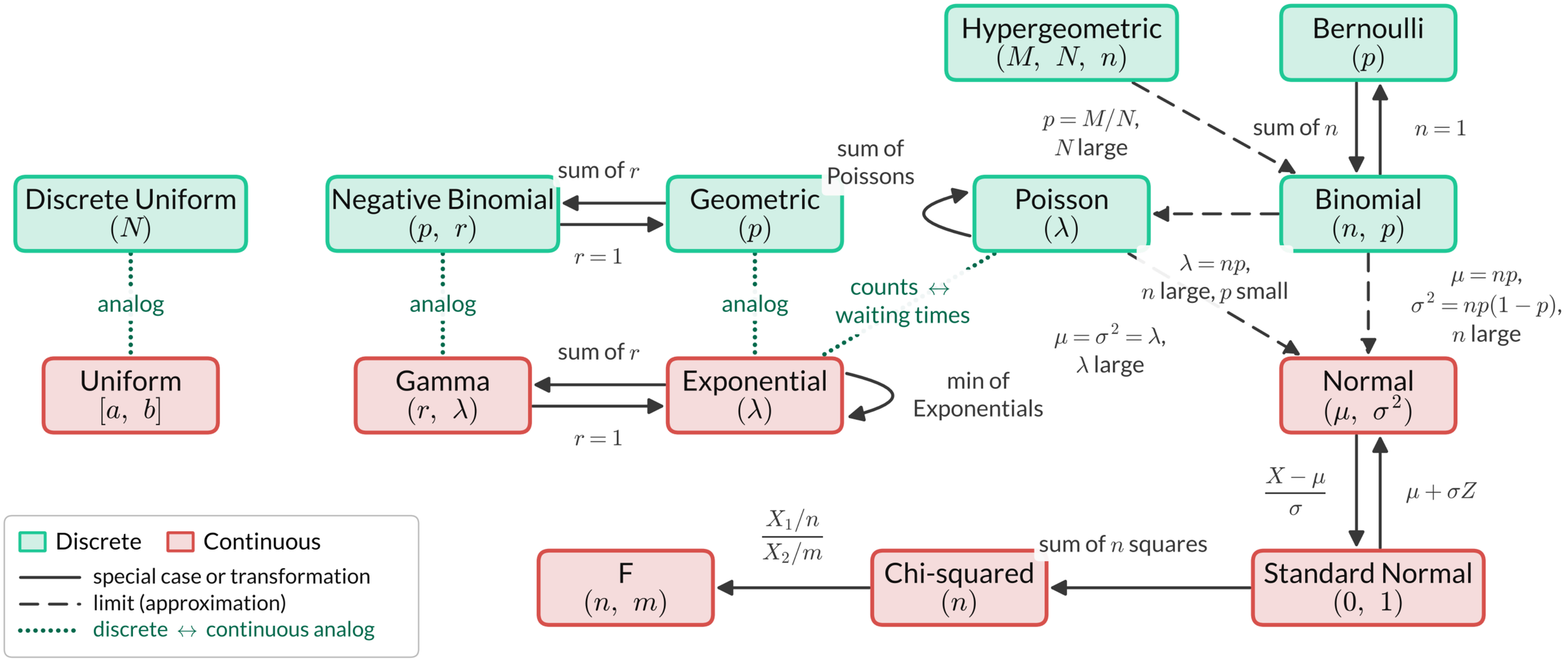
$$X = \sum_{i=1}^{n} Z_i^2$$

  has a Chi-squared distribution with n degrees of freedom
  - Shorthand: $X \sim \chi^2_{(n)}$
  - X is always positive (a sum of squares)
- A ratio of two independent chi-squared RVs, each divided by its degrees of freedom, has an F distribution: $\dfrac{\chi^2_{(n)}/n}{\chi^2_{(m)}/m} \sim F_{n,m}$



Relationships between distributions

Distributions and relationships we learned



Adapted from Leemis (1986)